Foundations of Principal component analysis

[Principal component analysis]

1.0 Content Block

Suppose you have data comprising a set of observations of p variables, and you want to reduce the data so that each observation can be described with only L variables, L < p. Suppose further, that the data are arranged as a set of n data vectors x 1 … x n {\displaystyle \mathbf {x} _{1}\ldots \mathbf {x} _{n}} with each x i {\displaystyle \mathbf {x} _{i}} representing a single grouped observation of the p variables.

2.0 Content Block

K-means clustering It has been asserted that the relaxed solution of k-means clustering, specified by the cluster indicators, is given by the principal components, and the PCA subspace spanned by the principal directions is identical to the cluster centroid subspace. However, that PCA is a useful relaxation of k-means clustering was not a new result, and it is straightforward to uncover counterexamples to the statement that the cluster centroid subspace is spanned by the principal directions.

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that is, that the data vector x {\displaystyle \mathbf {x} } is the sum of the desired information-bearing signal s {\displaystyle \mathbf {s} } and a noise signal n {\displaystyle \mathbf {n} } one can show that PCA can be optimal for dimensionality reduction, from an information-theoretic point-of-view. In particular, Linsker showed that if s {\displaystyle \mathbf {s} } is Gaussian and n {\displaystyle \mathbf {n} } is Gaussian noise with a covariance matrix proportional to the identity matrix, the PCA maximizes the mutual information I (y ; s) {\displaystyle I(\mathbf {y} ;\mathbf {s})} between the desired information s {\displaystyle \mathbf {s} } and the dimensionality-reduced output y = W L T x {\displaystyle \mathbf {y} =\mathbf {W} _{L}^{T}\mathbf {x} } . If the noise is still Gaussian and has a covariance matrix proportional to the identity matrix (that is, the components of the vector n {\displaystyle \mathbf {n} } are iid), but the information-bearing signal s {\displaystyle \mathbf {s} } is non-Gaussian (which is a common scenario), PCA at least minimizes an upper bound on the information loss, which is defined as

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History PCA was invented in 1901 by Karl Pearson, as an analogue of the principal axis theorem in mechanics; it was later independently developed and named by Harold Hotelling in the 1930s. Depending on the field of application, it is also named the discrete Karhunen–Loève transform (KLT) in signal processing, the Hotelling transform in multivariate quality control, proper orthogonal decomposition (POD) in mechanical engineering, singular value decomposition (SVD) of X (invented in the last quarter of the 19th century), eigenvalue decomposition (EVD) of XTX in linear algebra, factor analysis (for a discussion of the differences between PCA and factor analysis see Ch. 7 of Jolliffe's Principal Component Analysis), Eckart–Young theorem (Harman, 1960), or empirical orthogonal functions (EOF) in meteorological science (Lorenz, 1956), empirical eigenfunction decomposition (Sirovich, 1987), quasiharmonic modes (Brooks et al., 1988), spectral decomposition in noise and vibration, and empirical modal analysis in structural dynamics.

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X T X = W Σ T U T U Σ W T = W Σ T Σ W T = W Σ ^ 2 W T {\displaystyle {\begin{aligned}\mathbf {X} ^{T}\mathbf {X} &=\mathbf {W} \mathbf {\Sigma } ^{\mathsf {T}}\mathbf {U} ^{\mathsf {T}}\mathbf {U} \mathbf {\Sigma } \mathbf {W} ^{\mathsf {T}}\\&=\mathbf {W} \mathbf {\Sigma } ^{\mathsf {T}}\mathbf {\Sigma } \mathbf {W} ^{\mathsf {T}}\\&=\mathbf {W} \mathbf {\hat {\Sigma }} ^{2}\mathbf {W} ^{\mathsf

{T}}\end{aligned}}}

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Because the eigenvalues and eigenvectors of Σ (α) {\displaystyle \Sigma ^{(\alpha)}} are those of Σ {\displaystyle \Sigma } scaled by D {\displaystyle D} , the principal axes rotate toward any column whose variance has been inflated, exactly as the 2■D example below illustrates. If we have just two variables and they have the same sample variance and are completely correlated, then the PCA will entail a rotation by 45° and the "weights" (they are the cosines of rotation) for the two variables with respect to the principal component will be equal. But if we multiply all values of the first variable by 100, then the first principal component will be almost the same as that variable, with a small contribution from the other variable, whereas the second component will be almost aligned with the second original variable. This means that whenever the different variables have different units (like temperature and mass), PCA is a somewhat arbitrary method of analysis. (Different results would be obtained if one used Fahrenheit rather than Celsius for example.) Pearson's original paper was entitled "On Lines and Planes of Closest Fit to Systems of Points in Space" – "in space" implies physical Euclidean space where such concerns do not arise. One way of making the PCA less arbitrary is to use variables scaled so as to have unit variance, by standardizing the data and hence use the autocorrelation matrix instead of the autocovariance matrix as a basis for PCA. However, this compresses (or expands) the fluctuations in all dimensions of the signal space to unit variance. Classical PCA assumes the cloud of points has already been translated so its centroid is at the origin. Write each observation as

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Details PCA is defined as an orthogonal linear transformation on a real inner product space that transforms the data to a new coordinate system such that the greatest variance by some scalar projection of the data comes to lie on the first coordinate (called the first principal component), the second greatest variance on the second coordinate, and so on. Consider an n × p {\displaystyle n\times p} data matrix, X, with column-wise zero empirical mean (the sample mean of each column has been shifted to zero), where each of the n rows represents a different repetition of the experiment, and each of the p columns gives a particular kind of feature (say, the results from a particular sensor). Mathematically, the transformation is defined by a set of size l {\displaystyle l} (where l {\displaystyle l} is usually selected to be strictly less than p {\displaystyle p} to reduce dimensionality) of p {\displaystyle p} -dimensional vectors of weights or coefficients w (k) = (w 1 , … , w p) (k) {\displaystyle \mathbf {w} _{(k)}=(w_{1},\dots ,w_{p})_{(k)}} that map each row vector x (i) = (x 1 , … , x p) (i) {\displaystyle \mathbf {x} _{(i)}=(x_{1},\dots ,x_{p})_{(i)}} of X to a new vector of principal component scores t (i) = (t 1 , … , t l) (i) {\displaystyle \mathbf {t} _{(i)}=(t_{1},\dots ,t_{l})_{(i)}} , given by